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ORIGINAL PAPER

Deep Learning Auto-segmentation of Diffuse Midline Glioma on Multimodal Magnetic Resonance Images

Matías Fernández-Patón[1] · Alejandro Montoya-Filardi[2] · Adrián Galiana-Bordera[1] · Pedro Miguel Martínez-Gironés[1] · Diana Veiga-Canuto[1] · Blanca Martínez de las Heras[3] · Leonor Cerdá-Alberich[1] · Luis Martí-Bonmatí[1,2]

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Abstract

Diffuse midline glioma (DMG) H3 K27M–altered is a rare pediatric brainstem cancer with poor prognosis. To advance the development of predictive models to gain a deeper understanding of DMG, there is a crucial need for seamlessly integrating automatic and highly accurate tumor segmentation techniques. There is only one method that tries to solve this task in this cancer; for that reason, this study develops a modified CNN-based 3D-Unet tool to automatically segment DMG in an accurate way in magnetic resonance (MR) images. The dataset consisted of 52 DMG patients and 70 images, each with T1W and T2W or FLAIR images. Three different datasets were created: T1W images, T2W or FLAIR images, and a combined set of T1W and T2W/FLAIR images. Denoising, bias field correction, spatial resampling, and normalization were applied as preprocessing steps to the MR images. Patching techniques were also used to enlarge the dataset size. For tumor segmentation, a 3D U-Net architecture with residual blocks was used. The best results were obtained for the dataset composed of all T1W and T2W/FLAIR images, reaching an average Dice Similarity Coefficient (DSC) of 0.883 on the test dataset. These results are comparable to other brain tumor segmentation models and to state-of-the-art results in DMG segmentation using fewer sequences. Our results demonstrate the effectiveness of the proposed 3D U-Net architecture for DMG tumor segmentation. This advancement holds potential for enhancing the precision of diagnostic and predictive models in the context of this challenging pediatric cancer.

Keywords Segmentation · Pediatric oncology · Deep learningx · Diffuse midline glioma · Magnetic resonance

Introduction

Diffuse midline glioma H3 K27M–altered (DMG) formerly known as Diffuse Intrinsic Pontine Glioma (DIPG) is the most common brainstem pediatric cancer, although it is a rare entity with low incidence [1]. DMG is a highly aggressive and infiltrative brain tumor that originates from glial cells and arises in midline structures of the central nervous system, such as the pons, thalamus, and spinal cord. It is characterized by a specific histone mutation, H3 K27M, which plays a crucial role in tumor development and epigenetic dysregulation. This mutation has been recognized as a key molecular marker, leading to the reclassification of these tumors by the WHO in 2016 [2]. The lesion most frequently appears in children between 5 and 9 years old and has a notoriously poor prognosis, with a median survival time lower than a year from diagnosis [3, 4]. Due to its location and infiltrative nature, surgical resection is not a viable treatment option, while radiation and chemotherapy have shown limited efficacy [3].

In recent years, there has been a growing interest in exploring novel approaches to gain a deeper understanding of DMG. Clinical management of patients with DMG involves the use of MR at diagnosis and for follow-up assessments. Usually, the basic MR protocol includes, at

✉ Matías Fernández-Patón
matias_fernandez@iislafe.es

[1] Biomedical Imaging Research Group (GIBI230), Instituto de Investigación Sanitaria La Fe, Av. Fernando Abril Martorell, 106 Torre A, Planta 7ª, Despacho 7.22, Valencia 46026, Spain

[2] Department of Radiology, Hospital Universitario y Politécnico de La Fe, Valencia, Spain

[3] Department of Pediatric Oncology, Hospital Universitario y Politécnico de La Fe, Valencia, Spain

minimum, T1-weighted (T1 W), T2-weighted (T2 W), and/or Fluid Attenuated Inversion Recovery (FLAIR) images [5]. In T1 W images, the tumor appears hypointense with indistinct tumor margins, while in T2 W/FLAIR images, the tumor usually shows hyperintensity with sharp boundaries.

Artificial intelligence (AI) is increasingly recognized as a valuable tool in the classification, diagnosis, and management of DMG [6]. Advanced algorithms, including support vector machines (SVM) and CNN, have demonstrated high accuracy in differentiating DMGs from other brain tumors using MRI. Beyond structural analysis, AI is also being leveraged in the field of radiogenomics, enabling non-invasive prediction of genetic subtypes and informing personalized treatment strategies based on imaging characteristics. In parallel, prognostic models powered by AI have been developed to estimate overall and progression-free survival, supporting more informed clinical decision-making.

These advances in AI are closely connected with the field of radiomics, which involves the extraction and analysis of many quantitative features obtained from radiological images after the accurate definition of the Region of Interest (ROI). Radiomics has shown promising results across various cancer types, and recent studies have demonstrated its relevance in the context of DMG [7–10]. The objective of radiomics is to offer valuable insights into tumor characteristics, treatment response, and patient outcomes by leveraging advanced image processing techniques and machine learning algorithms.

To unlock the full potential of radiomics in tumor analysis, prior accurate tumor segmentation is essential. However, tumor segmentation remains a complex process that has not yet been integrated into clinical practice [11]. It requires manual intervention and expertise from trained radiologists, making it time-consuming, subjective, and prone to inter-observer variability [11, 12]. To streamline the process and improve consistency, several automated or semi-automated brain tumor segmentation methods have been developed [4, 11–28]. Automated methods offer the advantage of reducing human effort and variability, enabling faster and more reproducible segmentation. However, they may struggle with complex or heterogeneous tumors, requiring large, well-annotated harmonized datasets for training and validation. In contrast, semi-automated methods strike a balance between automation and expert oversight, enabling refinement that enhances accuracy. However, they still require a degree of manual intervention, which can be time-consuming and may introduce some variability.

Automatic deep learning-based segmentation aims to improve accuracy while significantly reducing the reliance on manual intervention—one of the most time-consuming and labor-intensive aspects of the segmentation process. Beyond its role in radiomics extraction, accurate tumor segmentation models are also crucial in treatment planning, preoperative assessment, and radiation therapy, where precise delineation of tumor boundaries can guide surgical decisions and optimize therapeutic interventions by reducing the inter-observer variability [11, 12]. This advancement holds substantial potential to improve the clinical management of DMG by enabling more precise tumor characterization and facilitating informed, treatment-guided decision-making.

Although deep learning has traditionally been applied to large image datasets, there is an increasing tendency to apply it to small datasets [29, 30]. To compensate for the lack of data, diverse types of network structures are used [29], such as networks already trained on other datasets [4] or data augmentation [30]. Similarly, multimodal analysis can improve the performance of image anomaly detection [31, 32]. In this way, the network can extract the most relevant information from each sequence, complementing the information that could be extracted if only one of the sequences were used, and achieving better results.

CNN-based models, specifically U-Nets architectures, have emerged as powerful tools for medical image segmentation, showcasing their ability to accurately delineate tumor regions [13, 16–18, 22, 24–28, 32]. This architecture exploits the encode-decode design with skip connection to combine context information and fine details, which also allows it to learn with relatively little data [23]. One of the most popular modifications of the U-Net is the ResU-Net [33], which incorporates the residual blocks of the ResNet [34] to improve the segmentation process. This architecture is widely used in the segmentation of glioma tumors in mono- [27] or multimodal images [26]. There are also architectures like this one with modifications such as the DFP-ResU-Net, a neural network designed for automatic segmentation of brain tumor subregions [29]. This net employs a spatial dilated feature pyramid (DFP) module, a residual module, and a U-Net structure. To the best of our knowledge, there are only a few studies [4, 35, 36] dealing with DMG segmentation. All of them use the same strategy, with a network already pre-trained on adult brain tumors, due to the small size of the cohort of DMG images. Two of them [35, 36] also used 4 sequences for this task: T1 W with and without contrast, FLAIR, and T2 W.

In rare diseases, such as DMG, applying data augmentation techniques becomes an essential requirement given the small size of the available datasets and the need to significantly increase the number of images with which we train. In the field of whole slide images, sample patching is widely used, not only to reduce the size of the images we work with, but also to increase the number of images with which the network is trained [37, 38]. This technique is not widely used in MRI but has been used in several cases [39, 40].

The main objective of this study was to create a robust automatic segmentation model using a CNN-based 3D U-Net architecture trained, validated, and tested on a small

multicenter real-world dataset of DMG patients from the Horizon 2020 PRIMAGE (Predictive In-silico Multiscale Analytics to support cancer personalized diagnosis and prognosis, Empowered by imaging biomarkers) project [41]. To achieve this goal, data augmentation techniques such as image patching, as well as 3 datasets constructed from T1 W images, T2 W/FLAIR images, and a combination of T1 W and T2 W/FLAIR images, are used to explore their performance.

Material and Methods

Datasets

The dataset used in this study was obtained as part of the Horizon 2020 PRIMAGE project [41]. A total of 52 children diagnosed with DMG were included. The diagnosis was made by MRI when the lesion originating from the pons exceeded 50% of its transverse diameter, appeared as a poorly defined mass with a diffuse growth pattern, hypointense on T1 W images, and hyperintense on T2 W images [42]. All of them possessed images at the time of diagnosis and 18 of them also after radiotherapy and/or chemotherapy treatment. The cases were provided mainly by a single center (22 cases from Hospital Universitario y Politécnico La Fe, Valencia, Spain) and, additionally, by external Spanish collaborators of the project (3 cases from Hospital Son Espases, Palma de Mallorca; 4 cases from Hospital Niño Jesus, Madrid; 11 cases from Hospital Virgen del Rocío, Sevilla; 7 cases from Hospital de Cruces, Barakaldo; and 5 cases from Hospital Reina Sofía, Córdoba). The patients' age at diagnosis was 7.5 ± 3.1 years (mean ± SD), ranging from 1.9 to 17 years, with a balanced gender distribution (24 girls and 28 boys). The inclusion criteria for this study were that patients were less than 21 years old and had an imaging diagnosis. Exclusion criteria excluded cases with no confirmed diagnosis, no imaging diagnosis at the time of diagnosis, or poor-quality images unsuitable for analysis.

All 70 MR exams had T1 W and T2 W (58) or FLAIR (12) MR sequences. The database exhibited substantial variability regarding field strength (53 1.5 T and 17 3 T), manufacturer (35 General Electric, 23 Philips, and 12 Siemens), in-plane voxel size (ranging from 0.21 to 1.02 mm) and slice thickness (from 0.8 to 5.5 mm).

Three datasets were then constituted: T1 W images, T2 W or FLAIR images (selecting FLAIR when available), and a combination of T1 W and T2 W or FLAIR images. The datasets were split into train, validation, and test sets ($n = 49$, $n = 7$, and $n = 14$) based on the manufacturer and center. To mitigate bias, patients with both diagnostic and post-treatment images had both sets of images placed within the same partition.

Manual Segmentation

A pediatric neuroradiologist with six years of experience in pediatric neuroradiology performed the manual segmentation by delineating and outlining the DMG tumor in the transversal plane in each slice. The most useful sequences to differentiate the tumor tissue from the adjacent non-tumoral parenchyma were the T2 W and FLAIR images in the transverse plane.

Any deviation from the normal signal intensity of the brainstem was included within the tumor area as minimal peritumoral edema is assumed in DMG and exclusion of potential tumor areas was minimized with this approach.

Image Preprocessing

MR images were harmonized to increase the reproducibility of the model and the quality of the images after applying denoising and bias field correction filters. Depending on the sequence, different denoising filters were applied based on image quality metrics previously published [43]. T1 W images underwent denoising using an anisotropic diffusion filter with a conductance of 0.5 and 1 iterations. In T2 W/FLAIR images, denoising was performed using a bilateral filter with a domain sigma of 0.5 and a range sigma of 60 using the SimpleITK library [44]. For all images, bias field correction was performed with N4 bias field correction of ANTS [45] with a B-spline of 50, a shrink factor of 2, and 50 and 30 iterations. T1 W images were registered to T2 W or FLAIR images using a rigid transformation with Elastix software [46]. Then, all images were rescaled to a voxel size of 1 × 1 × 1 mm, and the signal intensity was normalized using the z-score method. The steps of this process are shown in Fig. 1.

Data Augmentation

To increase the dataset's size, all training images were flipped along the sagittal plane and subsequently divided into patches of 64 × 64 × 64 voxels using the EMPatches library. Moreover, to further expand the dataset, all patches were overlapped by 50%. In this way, a more extensive spectrum of visual information was captured, aiming to enrich diversity and potentially enhance the performance of subsequent analysis or modeling tasks. These patches were selected to contain, at least, one tumor voxel. A total of 997 patches were captured (850 for training and 147 for validating), substantially increasing the number of patients with images for training (49 patients) and testing (7 patients). The 14 test image series were divided into 1826 patches. Unlike the training set, the test images were partitioned into patches irrespective of tumor presence, resulting in a substantially larger number of patches. This approach reflects

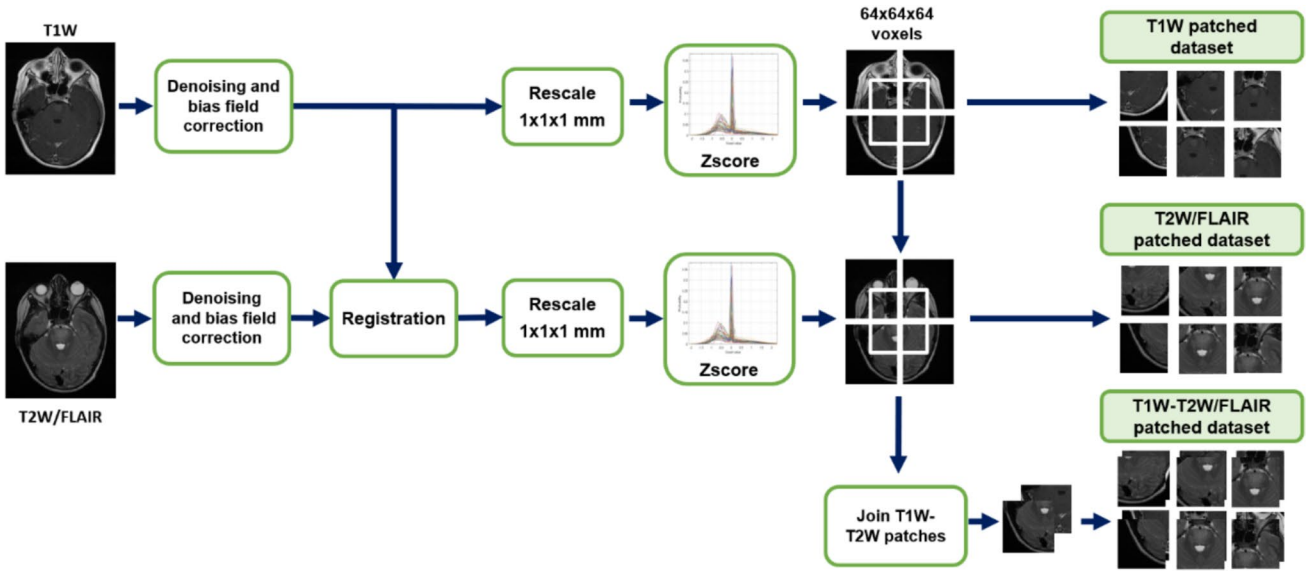


Fig. 1 Datasets preprocessing pipeline: registration, denoising bias field correction, rescale, normalization and patching

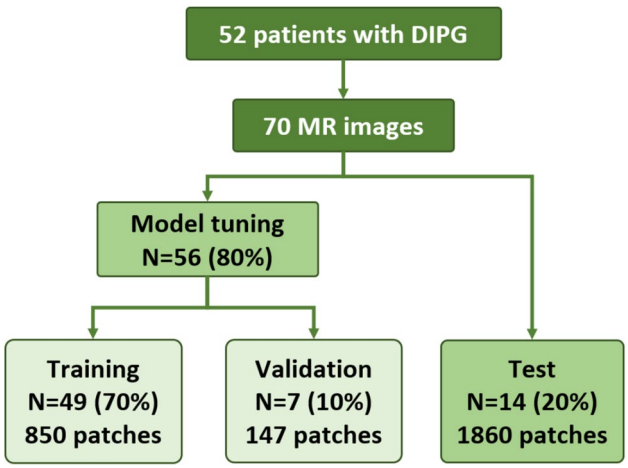


Fig. 2 DMG dataset distribution

the real-world inference scenario, where an entire volume is decomposed into patches and each patch is independently processed by the model, without prior knowledge of tumor localization. The segmented patches are then reassembled to reconstruct the full volume. The distribution of the data is shown in Fig. 2.

Segmentation Models

Three different segmentation models were trained, one for each dataset, to explore the advantages of using only T1 W, T2 W/FLAIR images or the combination of both sequences for the segmentation task. The selected architecture for training the DMG was an extension of the well-established 2D U-Net [23]. This deep learning solution is renowned for its effectiveness in medical image segmentation tasks. In this study, a modified 3D U-Net model will be used to adeptly manage and leverage the inherent three-dimensional nature of MR image volumes.

The fundamental concept behind the autoencoder 3D U-Net is the use of an encoder-decoder structure with skip connections. The encoder part consists of a series of convolutional layers that progressively reduce the spatial dimensions of the input while increasing the number of feature channels. This down sampling process captures hierarchical information and abstract features from the input volume. The first layers extract low-level features (edges, textures, gradients, etc.) that become more complex when moving to deeper layers. The decoder part, on the other hand, utilizes transposed convolutional layers to perform upsampling. These layers effectively increase the spatial resolution of the feature maps while decreasing the number of feature channels. Skip connections allow the gradients to flow directly from the encoder to the decoder, mitigating issues such as losing stuck information and helping in degradation problems in depth architectures.

To further enhance the performance of the model, residual blocks are added to the architecture. Residual blocks are placed between each convolution layer on both the encoder and decoder sides. This block, based on the Deep residual U-Nets approach [33], is composed of convolution, normalization, and activation layers, and preserves valuable information and facilitates the training of deeper networks, improving the information flow and reducing the problem of gradient decay. The key idea behind a residual block is its combination with skip connections. These connections allow

the network to learn residual mappings, capturing the difference between the input and the desired output. By adding the input directly to the output of the block, the network can learn to focus on the residual information, making it easier for the network to learn and optimize deeper layers.

The model input consists of a 3D volume of $64 \times 64 \times 64 \times N$, where N is the number of channels ($N = 1$ for T1 W or T2 W/FLAIR models and $N = 2$ for the combined T1 W and T2 W/FLAIR model). The output is a probability mask, where each voxel represents the probability of belonging to the tumor.

In Fig. 3, the final architecture and residual blocks layers can be observed.

Several hyperparameters were held constant, while others were subjected to tuning to determine the most optimal values. The number of epochs was set to 250, the learning rate was set to 0.0001, and the batch size was configured to 64 for the single sequence models and 32 for the combined sequence model. The final activation layer was chosen to be sigmoid, and the optimization algorithm employed was Adam [47].

The remaining hyperparameters, such as the choice of loss function (e.g., DSC [48] and Tversky [49]), intermediate activation layer (e.g., Relu [50] and Mish [51]) and the number of filters—referring to the number of convolution filters that are applied in the first convolutional layer—are presented in Table 1.

Mask Postprocessing

After segmenting all patches using the 3D U-Net model, the resulting masks were restored to their original dimensions through a process of assembling. The output probability masks were binarized using a threshold of 0.5. The

Table 1 Hyperparameters combinations for U-Net

Model	Loss function	Number of filters	Intermediate activation layer	Dropout
1	DSC	32	Relu	0
2	DSC	32	Relu	0.1
3	DSC	32	Relu	0.2
4	DSC	32	Mish	0
5	TVERSKY	32	Mish	0.1
6	TVERSKY	32	Relu	0.2
7	DSC	64	Relu	0
8	DSC	64	Relu	0.1

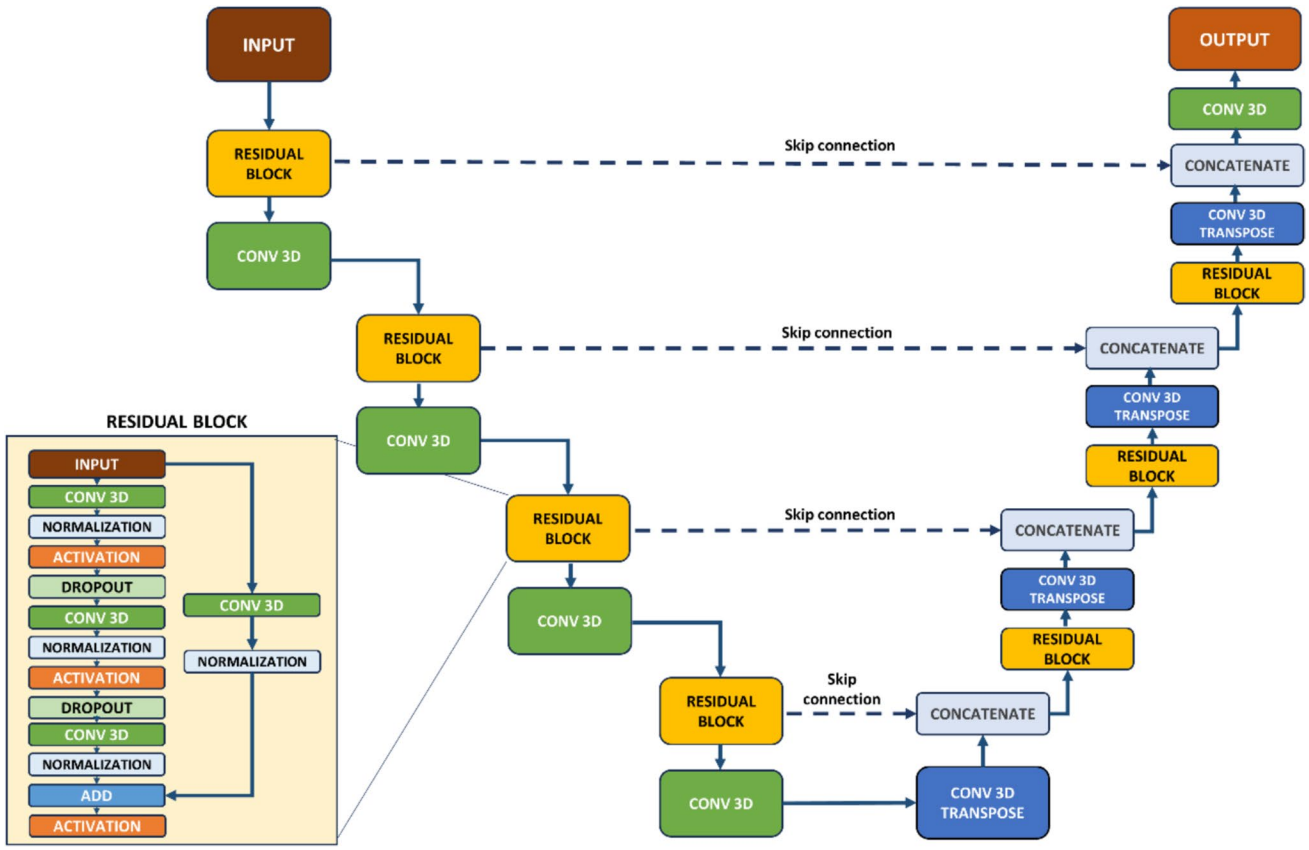


Fig. 3 Modified 3D U-Net architecture

morphological closing operation was then applied with a sphere of 1 voxel to smooth edges and fuse continuous regions. Finally, any 3D holes within the masks were filled given that the tumor is in essence regarded as a continuous mass devoid of holes.

Metrics

Considering the diverse nature of the metrics, multiple evaluation criteria were employed to thoroughly assess the model's performance. The following metrics were utilized: Dice Similarity Coefficient (DSC) [17–19, 21, 22, 52, 53], Jaccard index (JAC) [18, 21, 22, 52, 53], precision (Prec) [17], sensitivity (Sens) [17], and Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC) [53].

These metrics were calculated on the assembled tumor masks and post-processed. First, the image was divided into patches, regardless of whether they contained tumor or not. Each of these patches was introduced to the model for segmentation. Once all the patches were segmented, they were assembled to reconstruct the dimensions of the original image and post-processed. The complete workflow is shown in Fig. 4.

Experimental Setup

The proposed models have been trained and tested on an NVIDIA A100 80 GB Tensor Core GPU within the NVIDIA DGX A100 640 GB. The implemented models have been developed using the Keras library and TensorFlow. Once trained, the models have also been executed on the CPU of a laptop equipped with an Intel(R) Core(TM) i7-9750H CPU 2.60 GHz and 16 GB of RAM. The model execution code is accessible through the following link:

https://bitbucket.org/gibi230/preprocessing_segmentation_dipg/src/master/.

Results

The results for each dataset and model, both with and without mask postprocessing, are shown in Supplementary Tables 4, 5, and 6. Notably, the metrics associated with each parameter combination exhibit remarkable consistency, with marginal improvements being observed for certain configurations depending on the model. These 3D U-Net configurations for the best models are shown in Table 2.

Furthermore, the results obtained with the models combining T1 W and T2 W/FLAIR surpassed those achieved with T2 W/FLAIR alone, and, notably, these were superior to those obtained with the T1 W-only models. This trend becomes even more apparent in Table 3, which shows a summary of the best models for each dataset type after mask postprocessing. It can be seen for all the metrics. The T1 W models show the least favorable outcomes, followed by the T2 W model, and the combined sequence model emerges as the most proficient. Furthermore, a reduction in standard deviation is observable for models utilizing T2 W/FLAIR sequences.

The best results for the test dataset are shown in bold.

Figure 5 shows two examples of the tumor segmentation, both manual and as performed by each of the trained models across the various sequences. The figure provides a visual comparison between the manual mask, which serves as ground truth, and the automatic mask generated by each model. In the representations, true positives (TP) are indicated in green, false negatives (FN) in yellow, and false positives (FP) in red.

The total training execution time was 7.88 ± 5.53 h. The execution time of the complete pipeline from denoising to final mask reconstruction for each patient was 10.25 ± 4.43 min on the laptop and 2.39 ± 1.35 min on the NVIDIA DGX A100.

Table 2 3D U-Net hyperparameters configuration for each dataset

Hyperparameter	T1 W	T2 W/FLAIR	T1-T2 W/FLAIR
Epochs	250	250	250
Batch size	64	64	32
Dropout	0.1	0	0
Loss function	DSC	DSC	DSC
Learning rate	0.0001	0.0001	0.0001
No. of filter	64	32	64
Intermediate activation layer	Relu	Mish	Relu
Final activation layer	Sigmoid	Sigmoid	Sigmoid
Optimization	Adam	Adam	Adam

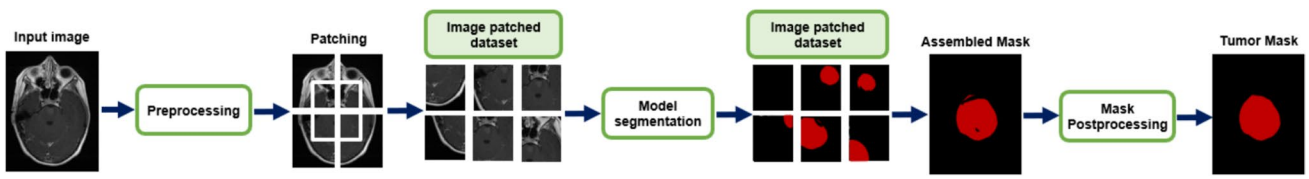


Fig. 4 Segmentation workflow

Table 3 Segmentations metrics results (mean ± standard deviation)

Model	Dataset	DSC	JAC	AUC	Prec	Sens
T1 W	**Train**	0.931 ± 0.031	0.873 ± 0.055	0.968 ± 0.022	0.928 ± 0.039	0.936 ± 0.032
	Validation	0.816 ± 0.092	0.698 ± 0.134	0.886 ± 0.071	0.883 ± 0.121	0.773 ± 0.131
	Test	0.863 ± 0.058	0.764 ± 0.102	0.912 ± 0.052	0.889 ± 0.082	0.824 ± 0.109
T2 W/FLAIR	**Train**	0.935 ± 0.019	0.881 ± 0.043	0.961 ± 0.023	0.938 ± 0.027	0.923 ± 0.042
	Validation	0.883 ± 0.047	0.792 ± 0.068	0.927 ± 0.049	0.896 ± 0.093	0.855 ± 0.093
	Test	0.868 ± 0.048	0.769 ± 0.068	0.928 ± 0.037	0.909 ± 0.056	0.856 ± 0.078
T1 W-T2 W/FLAIR	**Train**	0.940 ± 0.031	0.889 ± 0.047	0.972 ± 0.021	0.949 ± 0.018	0.944 ± 0.026
	Validation	0.857 ± 0.051	0.753 ± 0.082	0.915 ± 0.042	0.916 ± 0.047	0.831 ± 0.079
	Test	**0.883 ± 0.052**	**0.793 ± 0.082**	**0.933 ± 0.043**	**0.921 ± 0.048**	**0.866 ± 0.089**

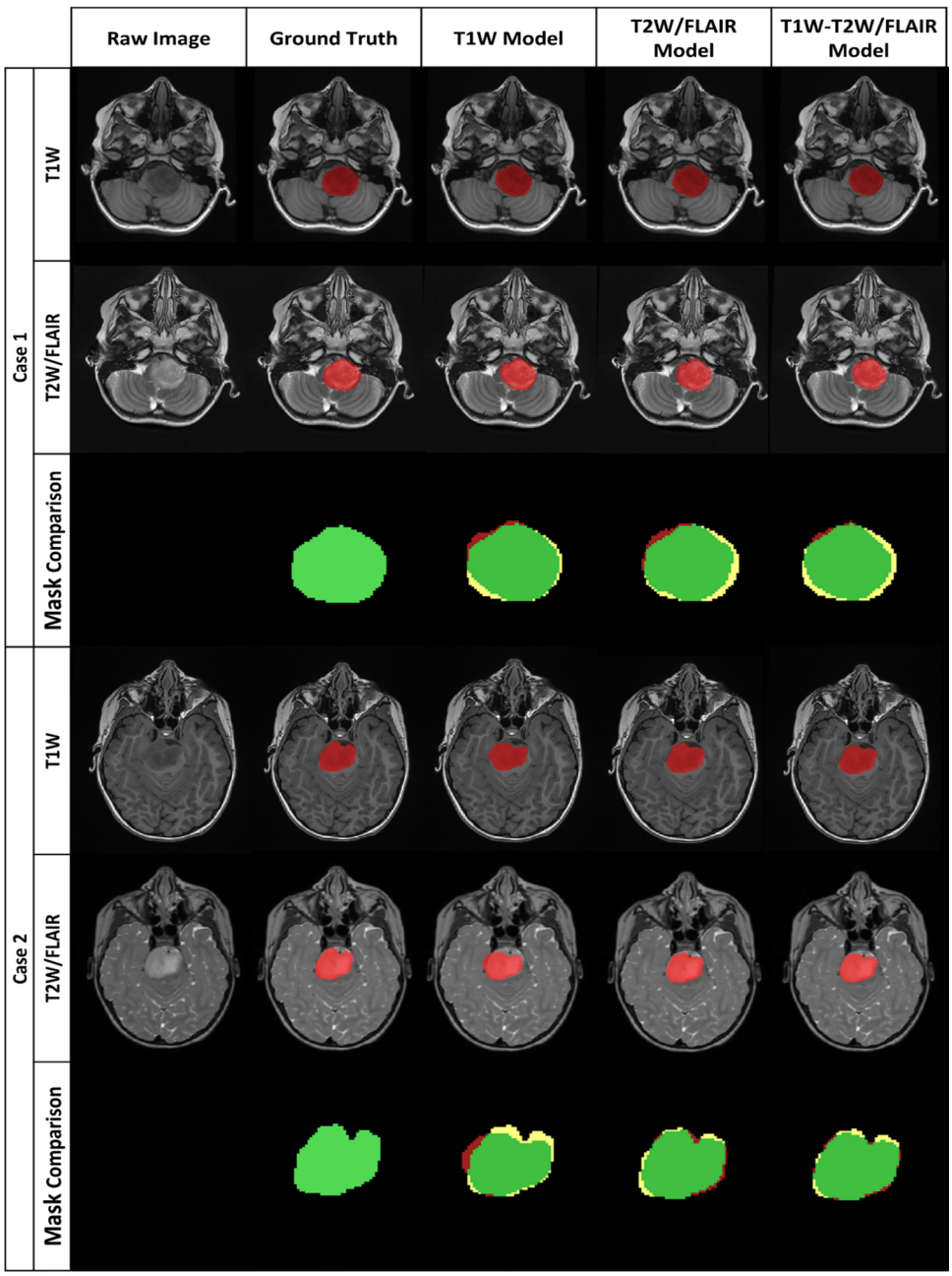


Fig. 5 Two examples of manual and automatic segmentation for each model and the comparison of both manual and automatic segmentation masks (TP: green, FN: yellow, and FP: red)

Discussion

Our study explores the performance of a modified 3D U-Net architecture with additional residual blocks for the automatic segmentation of DMG tumors. This architecture was tested with different hyperparameters in three different datasets to reach the best results: T1 W dataset, T2 W/FLAIR dataset, and a merged dataset combining the T1 W and T2 W/FLAIR sequences. These datasets were composed of 70 MR multicenter imaging studies that were split into training, validation, and test sets. To validate the efficacy of the modified 3D U-Net architecture, the trained model was applied to the test dataset for DMG tumor segmentation. This test dataset exclusively contained images that were not used during the training and fine-tuning phases.

To accelerate the learning process, the images were preprocessed due to the multicentric variability of the images. The application of denoising adapted to each modality, bias field correction, resampling, and normalization not only improves model learning but also helps to improve model reproducibility by harmonizing the model input images.

Data augmentation techniques such as image patching have been used. The input to the network is a patch instead of a complete image, which greatly increases the number of images in the ensemble. In this way, it has been possible to deal with this type of problem where the number of available cases is very small.

The T1 W utilized twice the number of filters compared to the T2 W/FLAIR model, with 64 and 32, respectively. This could be due to the segmentation task being more challenging when employing T1 W, as the boundaries are less well-defined. This can also be shown with the best T1 W model as it is the only one that employs a dropout layer; i.e., it is the only one that needs this layer to prevent overfitting. The rest of the models generalize well in the test database without the need for this layer to prevent overfitting. Overall, the results demonstrate that models constructed with T2 W or FLAIR images exhibit slightly better performance than the model based on T1 W images. Furthermore, the combination of T1 W and T2 W or FLAIR images enhances performance compared to using only T2 W/FLAIR or T1 W images. This is in line with the results obtained from other brain studies [31, 32] and specifically in DMG [35, 36], in which the integration of information from different image sequences appears to provide an additional advantage, suggesting that combining multiple image sequences can lead to more accurate DMG automatic segmentation.

These same results can also be observed in Fig. 5, where the model that uses both sequences gets the fewest false positives (FP), as can be seen in the Precision column in Table 2. Most errors are concentrated along the edges. DMG is a highly diffuse tumor with poorly defined and diffused boundaries, which can pose challenges for the network in segmenting the edges. When we focus on segmenting only the tumor, we prefer to focus only on tumor tissue, excluding any surrounding non-tumor tissue. This idea is also mentioned in the literature [53], where the inclusion of false positives has a much greater impact than false negatives due to the introduction of significant bias in the extraction of radiomic features for the development of predictive models. By doing so, we ensure that everything included is indeed tumor tissue. In Supplementary Tables 4, 5, and 6, it is also observed that postprocessing of automatic mask segmentation led to an improvement in the performance of the final segmentation.

Additionally, it was observed that, for certain models, the performance metrics on the test set were slightly higher than those on the validation set. This discrepancy may be attributed to the smaller size of the validation partition, which could include more challenging cases for segmentation, thereby lowering average performance. In contrast, the test set encompasses a broader and more heterogeneous range of cases, potentially diluting the impact of particularly difficult DMG examples and resulting in slightly improved overall metrics.

Comparing our results with the state of the art for the segmentation of other gliomas, the performance of our model is like this, even superior. These studies use the database published with more than 1200 images: BraTS 2015 obtaining a DSC of 0.83 [27], BraTS 2019 with a DSC of 0.90 [25], and BraTS 2020 of 0.81 [26]. Moreover, three published articles focused on automatic segmentation in DMG [4, 35, 36]. In these articles, they use a model trained on brain tumors to detect rare tumors such as DMG. The first model [4] used T2 W and/or FLAIR images and achieved a maximum DSC of 0.622 when using FLAIR images on a test dataset consisting of 62 images. The second model [35] achieved a DSC of 0.88 in the internal cohort of 53 images and 0.86 in the external cohort of 16 images using T1 W with and without contrast, T2 W, and FLAIR. The last model [36] obtained a mean DSC of 0.880 in 9 patients using the same MR sequences as the previous model. Our models achieved significantly higher DSC values for all test datasets (0.863 for T1 W, 0.868 for T2 W/FLAIR, and 0.883 for T1 W-T2 W/FLAIR model) compared to the first model and showed comparable performance to the other two. This demonstrates the power and robustness of the model used and the advantage of using image patching to increase the number of images, with results comparable to other studies, with the main advantage of using fewer MR sequences (T1 W and T2 W/FLAIR). This is a great advantage as not all studies have FLAIR and contrast-enhanced T1 W images.

In addition, it has been found that the execution times on a basic laptop can be assumed. This supports the implementation of the algorithm in clinical practice by ensuring an efficient and practical application in clinical settings.

The main limitation of our study is the size of the dataset, a constraint acknowledged in the same publication [2]. Addressing this problem, data augmentation has been applied in two diverse ways. The first is by flipping the images, and the second and more important is by dividing each image into patches. This allows a considerable increase in the number of images from which the net can learn more effectively. Despite this, only 14 of these images were used to test the model performance. Further external prospective validation studies are warranted to evaluate the accuracy of this tool in diverse populations. It also has to be considered that many of the images come from a single center, ideally involving at least two radiologists to assess interobserver variability in comparison with the model. Additionally, a thorough investigation of the potential benefits and limitations of its clinical application is essential to support its integration into routine practice.

Conclusion

In conclusion, a modified 3D-Unet capable of detecting and segmenting DMGs automatically and accurately using data augmentation techniques such as patching has been developed from a small multicenter dataset. The best model combines T1 W and T2 W/FLAIR images, providing a multimodal perspective without the mandatory use of sequences such as contrast enhanced T1 W or FLAIR images, which are not always available. This tool not only provides sufficient accuracy, efficiency, and reproducibility for clinical segmentation of DMG tumors but also serves as a direct decision support tool for radiologists by directly providing the tumor region, thus saving them from this time-consuming and laborious task. This advance greatly facilitates the wider adoption of radiomics in clinical practice and the development of predictive models that could, as a last step, help to better understand clinical evolution, especially in this rare and visually complicated tumor.

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Declarations

Ethics approval This study has been approved by the Hospital's Ethics Committee (The Ethics Committee for Investigation with medicinal products of the University and Polytechnic La Fe Hospital, ethic code: 2018/0228).

The research conducted respects fundamental ethical principles, including those reflected in the Charter of Fundamental Rights of the European Union (7 th December 2000) and worldwide: Dignity, Freedom, Equality, Solidarity, Citizens' Rights and Justice. Additionally, the project is in accordance with the European Human Rights Convention, especially with regard to Privacy and Autonomy. PRIMAGE project did not work directly with any human being, as only patients' data was handled. The research did not involve physical intervention in the study's participants. The research did not involve collection of biological samples and as patients were already treated. PRIMAGE project has the favorable dictum of the Ethics Committee Research from the University and Polytechnic Hospital La Fe, the St. Anna Kinderkrebsforschung Children's Cancer Research Institute (CCRI), the University Clinic of Koeln, and the approval of the AEMPS (Spanish Agency of Medicine and Health products) and the University Hospital of Pisa (UNIPI).

Competing interests The authors declare no competing interests.

References

1. Jovanovich, N., Habib, A., Head, J., Hameed, F., Agnihotri, S. and Zinn, P.O.: Pediatric diffuse midline glioma: Understanding the mechanisms and assessing the next generation of personalized therapeutics. *Neuro-Oncology Advances*,. https://doi.org/10.1093/noajnl/vdad040, 2023.
2. Louis, D.N., Perry, A., Reifenberger, G., Von Deimling, A., Figarella-Branger, D., Cavenee, W.K. et al.: The 2016 World Health Organization classification of tumors of the central nervous system: a summary. *Acta Neuropathologica*, : Springer. **131**, 803–20. , 2016.
3. Vanan, M.I. and Eisenstat, D.D.: DIPG in children - What can we learn from the past? *Frontiers in Oncology*, **5**. https://doi.org/10.3389/fonc.2015.00237, 2015.
4. Chegraoui, H., Philippe, C., Dangouloff-Ros, V., Grigis, A., Calmon, R., Boddaert, N. et al.: Object detection improves tumour segmentation in mr images of rare brain tumours. *Cancers*, **13**, 1–19. https://doi.org/10.3390/cancers13236113, 2021.
5. Huang, R.Y. and Wen, P.Y.: Response Assessment in Neuro-Oncology Criteria and Clinical Endpoints. *Magnetic Resonance Imaging Clinics of North America*, : Elsevier Inc. **24**, 705–18. https://doi.org/10.1016/j.mric.2016.06.003, 2016.
6. Avval, A.H., Banerjee, S., Zielke, J., Kann, B.H., Mueller, S. and Rauschecker, A.M.: Applications of artificial intelligence and advanced imaging in pediatric diffuse midline glioma. 1–15. , 2025.
7. Wagner, M.W., Namdar, K., Napoleone, M., Hainc, N., Amirabadi, A., Fonseca, A. et al.: Radiomic Features Based on MRI Predict Progression-Free Survival in Pediatric Diffuse Midline Glioma/Diffuse Intrinsic Pontine Glioma. *Canadian Association of Radiologists Journal*,. https://doi.org/10.1177/08465371221109921, 2023.
8. Tam, L.T., Yeom, K.W., Wright, J.N., Jaju, A., Radmanesh, A., Han, M. et al.: MRI-based radiomics for prognosis of pediatric diffuse intrinsic pontine glioma: An international study. *Neuro-Oncology Advances*, **3**, 1–9. https://doi.org/10.1093/noajnl/vdab042, 2021.

9. Goya-Outi, J., Calmon, R., Orlhac, F., Philippe, C., Boddaert, N., Puget, S. et al.: Can structural MRI radiomics predict DIPG histone H3 mutation and patient overall survival at diagnosis time? *2019 IEEE EMBS International Conference on Biomedical and Health Informatics, BHI 2019 - Proceedings*,. https://doi.org/10.1109/BHI.2019.8834524, 2019.
10. Tinkle, C., Hsu, C.-Y., Simpson, E., Chiang, J., Li, X., Armstrong, J. et al.: NIMG-51. CONVENTIONAL MRI RADIOMIC FEATURES IMPROVE PROGNOSTICATION AND ARE PREDICTIVE OF H3 K27M STATUS IN DIPG. *Neuro-Oncology*,. https://doi.org/10.1093/neuonc/noaa215.664, 2020.
11. Işin, A., Direkoğlu, C. and Şah, M.: Review of MRI-based Brain Tumor Image Segmentation Using Deep Learning Methods. *Procedia Computer Science*, **102**, 317–24. https://doi.org/10.1016/j.procs.2016.09.407, 2016.
12. Munir, K., Frezza, F. and Rizzi, A.: Deep learning for brain tumor segmentation. *Studies in Computational Intelligence*, **908**, 189–201. https://doi.org/10.1007/978-981-15-6321-8_11, 2021.
13. Thillaikkarasi, R. and Saravanan, S.: An Enhancement of Deep Learning Algorithm for Brain Tumor Segmentation Using Kernel Based CNN with M-SVM. *Journal of Medical Systems*,. https://doi.org/10.1007/s10916-019-1223-7, 2019.
14. Zhou, Z., He, Z. and Jia, Y.: AFPNet: A 3D fully convolutional neural network with atrous-convolution feature pyramid for brain tumor segmentation via MRI images. *Neurocomputing*,. https://doi.org/10.1016/j.neucom.2020.03.097, 2020.
15. Khan, H., Shah, P.M., Shah, M.A., Islam, S. ul and Rodrigues, J.J.P.C.: Cascading handcrafted features and Convolutional Neural Network for IoT-enabled brain tumor segmentation. *Computer Communications*,. https://doi.org/10.1016/j.comcom.2020.01.013, 2020.
16. Qamar, S., Jin, H., Zheng, R., Ahmad, P. and Usama, M.: A variant form of 3D-UNet for infant brain segmentation. *Future Generation Computer Systems*, **108**. https://doi.org/10.1016/j.future.2019.11.021, 2020.
17. Chen, W., Liu, B., Peng, S., Sun, J. and Qiao, X.: S3D-UNET: Separable 3D U-Net for brain tumor segmentation. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*,. https://doi.org/10.1007/978-3-030-11726-9_32, 2019.
18. Jia, Q. and Shu, H.: BiTr-Unet: A CNN-Transformer Combined Network for MRI Brain Tumor Segmentation. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*,. https://doi.org/10.1007/978-3-031-09002-8_1, 2022.
19. Agravat, R.R. and Raval, M.S.: Brain tumor segmentation and survival prediction. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*,. https://doi.org/10.1007/978-3-030-46640-4_32, 2020.
20. Balafar, M.A., Ramli, A.R., Saripan, M.I. and Mashohor, S.: Review of brain MRI image segmentation methods [Internet]. Artif Intell Rev. : Springer. p. 261–74. https://doi.org/10.1007/s10462-010-9155-0, 2010.
21. Supraja, T., Sulthana, T.R., Pilani, S. and Veeramakali, D.C.: Brain Tumor Segmentation and Prediction using Fuzzy Neighborhood Learning Approach for 3D MRI Images. 0–16. , 2021.
22. Ozgur Cicek, Ahmed Abdulkabdir, Soeren S. Lienkamp, Thomas Brox, O.R.: 3D U_net. *Medical Image Computing and Computer-Assisted Intervention*, 424–32. https://doi.org/10.1007/978-3-319-46723-8, 2016.
23. Ronneberger, O., Fischer, P. and Brox, T.: U-net: Convolutional networks for biomedical image segmentation. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*,. https://doi.org/10.1007/978-3-319-24574-4_28, 2015.
24. Li, X., Jiang, Y., Li, M., Zhang, J., Yin, S. and Luo, H.: MSFR-Net: Multi-modality and single-modality feature recalibration network for brain tumor segmentation. *Medical Physics*, **50**. https://doi.org/10.1002/mp.15933, 2023.
25. Wang, J., Gao, J., Ren, J., Luan, Z., Yu, Z., Zhao, Y. et al.: DFP-ResUNet:Convolutional Neural Network with a Dilated Convolutional Feature Pyramid for Multimodal Brain Tumor Segmentation. *Computer Methods and Programs in Biomedicine*,. https://doi.org/10.1016/j.cmpb.2021.106208, 2021.
26. Kalaiselvi, T. and Padmapriya, S.T.: Multimodal MRI Brain Tumor Segmentation-A ResNet-based U-Net approach. *Brain Tumor MRI Image Segmentation Using Deep Learning Techniques*,. https://doi.org/10.1016/B978-0-323-91171-9.00013-2, 2021.
27. Shehab, L.H., Fahmy, O.M., Gasser, S.M. and El-Mahallawy, M.S.: An efficient brain tumor image segmentation based on deep residual networks (ResNets). Journal of King Saud University - Engineering Sciences. https://doi.org/10.1016/j.jksues.2020.06.001, 2021.
28. Yaqub, M., Jinchao, F., Zia, M.S., Arshid, K., Jia, K., Rehman, Z.U. et al.: State-of-the-art CNN optimizer for brain tumor segmentation in magnetic resonance images. *Brain Sciences*, **10**, 1–19. https://doi.org/10.3390/brainsci10070427, 2020.
29. Brigato, L. and Iocchi, L.: A close look at deep learning with small data. *Proceedings - International Conference on Pattern Recognition*, 2490–7. https://doi.org/10.1109/ICPR48806.2021.9412492, 2020.
30. Goceri, E.: Medical image data augmentation: techniques, comparisons and interpretations [Internet]. Artif Intell Rev. : Springer Netherlands. https://doi.org/10.1007/s10462-023-10453-z, 2023.
31. Rashid, T., Liu, H., Ware, J.B., Li, K., Romero, J.R., Fadaee, E. et al.: Deep learning based detection of enlarged perivascular spaces on brain MRI. *Neuroimage: Reports*, **3**, 1–29. https://doi.org/10.1016/j.ynirp.2023.100162, 2023.
32. Beser-Robles, M., Castellá-Malonda, J., Martínez-Gironés, P.M., Galiana-Bordera, A., Ferrer-Lozano, J., Ribas-Despuig, G. et al.: Deep learning automatic semantic segmentation of glioblastoma multiforme regions on multimodal magnetic resonance images. *International Journal of Computer Assisted Radiology and Surgery*, **19**, 1743–51. https://doi.org/10.1007/s11548-024-03205-z, 2024.
33. Zhang, Z., Liu, Q. and Wang, Y.: Road Extraction by Deep Residual U-Net. *IEEE Geoscience and Remote Sensing Letters*, **15**, 749–53. https://doi.org/10.1109/LGRS.2018.2802944, 2018.
34. He, K., Zhang, X., Ren, S. and Sun, J.: Deep residual learning for image recognition. *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*,. https://doi.org/10.1109/CVPR.2016.90, 2016.
35. Liu, X., Jiang, Z., Roth, H.R., Anwar, S.M., Bonner, E.R., Mahtabfar, A. et al.: Early prognostication of overall survival for pediatric diffuse midline gliomas using MRI radiomics and machine learning: A two-center study. *Neuro-Oncology Advances*, **6**, 1–11. https://doi.org/10.1093/noajnl/vdae108, 2024.
36. Liu, X., Bonner, E.R., Jiang, Z., Roth, H.R., Muhammad Anwar, S., Packer, R.J. et al.: Automatic Segmentation of Rare Pediatric Brain Tumors Using Knowledge Transfer From Adult Data. *Proceedings - International Symposium on Biomedical Imaging*, **2023-April**. https://doi.org/10.1109/ISBI53787.2023.10230757, 2023.
37. Khened, M., Kori, A., Rajkumar, H., Krishnamurthi, G. and Srinivasan, B.: A generalized deep learning framework for whole-slide image segmentation and analysis. *Scientific Reports*, : Nature Publishing Group UK. **11**, 1–14. https://doi.org/10.1038/s41598-021-90444-8, 2021.

38. Berman, A.G., Orchard, W.R., Gehrung, M. and Markowetz, F.: SliDL: A toolbox for processing whole-slide images in deep learning. *PLoS ONE*, **18**, 1–25. https://doi.org/10.1371/journal.pone.0289499, 2023.
39. Zhang, X., Han, L., Han, L., Chen, H., Dancey, D. and Zhang, D.: sMRI-PatchNet: A Novel Efficient Explainable Patch-Based Deep Learning Network for Alzheimer's Disease Diagnosis With Structural MRI. *IEEE Access*, **11**, 108603–16. https://doi.org/10.1109/ACCESS.2023.3321220, 2023.
40. Yalcinkaya, D.M., Youssef, K., Heydari, B., Zamudio, L., Dharmakumar, R. and Sharif, B.: Deep Learning-Based Segmentation and Uncertainty Assessment for Automated Analysis of Myocardial Perfusion MRI Datasets Using Patch-Level Training and Advanced Data Augmentation. *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBS*, 4072–8. https://doi.org/10.1109/EMBC46164.2021.9629581, 2021.
41. Martí-Bonmatí, L., Alberich-Bayarri, Á., Ladenstein, R., Blanquer, I., Segrelles, J.D., Cerdá-Alberich, L. et al.: PRIMAGE project: predictive in silico multiscale analytics to support childhood cancer personalised evaluation empowered by imaging biomarkers. *European Radiology Experimental*,. https://doi.org/10.1186/s41747-020-00150-9, 2020.
42. Barkovich, A.J., Krischer, J., Kun, L.F., Packer, R., Zimmerman, R.A., Freeman, C.R. et al.: Brain stem gliomas: A classification system based on magnetic resonance imaging. *Pediatric Neurosurgery*,. https://doi.org/10.1159/000120511, 1990.
43. Fernández Patón, M., Cerdá Alberich, L., Sangüesa Nebot, C., Martínez de las Heras, B., Veiga Canuto, D., Cañete Nieto, A. et al.: MR Denoising Increases Radiomic Biomarker Precision and Reproducibility in Oncologic Imaging. *Journal of Digital Imaging*, : Springer International Publishing. **34**, 1134–45. https://doi.org/10.1007/s10278-021-00512-8, 2021.
44. Beare, R., Lowekamp, B. and Yaniv, Z.: Image segmentation, registration and characterization in R with simpleITK. *Journal of Statistical Software*,. https://doi.org/10.18637/jss.v086.i08, 2018.
45. Tustison, N.J., Avants, B.B., Cook, P.A., Zheng, Y., Egan, A., Yushkevich, P.A. et al.: N4ITK: Improved N3 bias correction. *IEEE Transactions on Medical Imaging*,. https://doi.org/10.1109/TMI.2010.2046908, 2010.
46. Klein, S., Staring, M., Murphy, K., Viergever, M.A. and Pluim, J.P.W.: elastix: A Toolbox for Intensity-Based Medical Image Registration. *IEEE TRANSACTIONS ON MEDICAL IMAGING*, **29**. https://doi.org/10.1109/TMI.2009.2035616, 2010.
47. Kingma, D.P. and Ba, J.L.: Adam: A method for stochastic optimization. *3rd International Conference on Learning Representations, ICLR 2015 - Conference Track Proceedings*, 1–15. , 2015.
48. Milletari, F., Navab, N. and Ahmadi, S.A.: V-Net: Fully convolutional neural networks for volumetric medical image segmentation. *Proceedings - 2016 4th International Conference on 3D Vision, 3DV 2016*,. https://doi.org/10.1109/3DV.2016.79, 2016.
49. Salehi, S.S.M., Erdogmus, D. and Gholipour, A.: Tversky loss function for image segmentation using 3D fully convolutional deep networks. *Lecture Notes in Computer Science (Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, **10541 LNCS**, 379–87. https://doi.org/10.1007/978-3-319-67389-9_44, 2017.
50. Nair, V. and Hinton, G.E.: Rectified linear units improve Restricted Boltzmann machines. *ICML 2010 - Proceedings, 27th International Conference on Machine Learning*,. , 2010.
51. Misra, D.: Mish: A Self Regularized Non-Monotonic Activation Function. , 2019.
52. Veiga-Canuto, D., Cerdà-Alberich, L., Jiménez-Pastor, A., Carot Sierra, J.M., Gomis-Maya, A., Sangüesa-Nebot, C. et al.: Independent Validation of a Deep Learning nnU-Net Tool for Neuroblastoma Detection and Segmentation in MR Images. *Cancers*, **15**, 1–13. 2023, .https://doi.org/10.3390/cancers15051622
53. Veiga-Canuto, D., Cerdà-Alberich, L., Sangüesa Nebot, C., Martínez de las Heras, B., Pötschger, U., Gabelloni, M. et al.: Comparative Multicentric Evaluation of Inter-Observer Variability in Manual and Automatic Segmentation of Neuroblastic Tumors in Magnetic Resonance Images. *Cancers*, **14**. https://doi.org/10.3390/cancers14153648, 2022.

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